

1. Administrative & Study Identification

Full Study Title: A PHASE 3, EXTERNAL AND SYNTHETIC PLACEBO-CONTROLLED RANDOMIZED STUDY WITH DOSE-UP FOR NON-RESPONDERS TO INVESTIGATE SAFETY AND EFFICACY OF RITLECITINIB 50 MG AND 100 MG ONCE DAILY IN ADULT AND ADOLESCENT PARTICIPANTS 12 YEARS OF AGE AND OLDER WITH ALOPECIA AREATA **Short Title/Acronym:** A Study of 2 Doses of Ritlecitinib in People 12 Years of Age and Older With Alopecia Areata **Protocol Number:** B7981094 (Secondary ID: 2024-519370-40-00) **NCT Number:** NCT06873945 **Posting ID:** 674030387309581743 **Sponsor Name:** Pfizer **Investigational Product:** Ritlecitinib (Trade name: Litfulo) (50 mg and 100 mg) **Phase of Development:** Phase 3 **Trial Status:** Recruiting **Medical Monitor:** Pfizer Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate whether ritlecitinib 100 mg once daily (QD) provides superior scalp hair regrowth compared with the approved 50 mg dose and an external placebo control group in participants with severe alopecia areata. **Secondary Objectives:** To evaluate the efficacy and safety of ritlecitinib 100 mg versus a synthetic placebo control at Week 36; ritlecitinib 100 mg versus 50 mg among Week 24 non-responders at Week 48; and the change from baseline in SALT (Severity of Alopecia Tool) score at Week 24.

2.2 Background & Rationale Despite recent therapeutic advancements, current treatments for severe alopecia areata (AA) demonstrate limited efficacy. Across major JAK inhibitor trials, 70-87% of patients with severe AA do not achieve a SALT score ≤ 20 at week 24 on approved doses. Pharmacokinetic and exposure-response modeling suggests that a higher dose (100 mg) of ritlecitinib may yield additional clinical benefits. This study utilizes an innovative methodology that eliminates direct placebo randomization by incorporating external and synthetic placebo control arms derived from prior ritlecitinib studies, maintaining ethical continuity of care for severe AA patients.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** External and synthetic placebo-controlled **Blinding:** Double-blind **Randomization:** Randomized (participants are initially assigned by chance to 50 mg or 100 mg. Week 24 non-responders on 50 mg are

re-randomized 2:1 to either escalate to 100 mg or continue 50 mg).

3.2 Planned Enrollment & Duration Target **\$N\$:** Approximately 550 participants **Number of Sites:** Multicenter global study (including locations in the United States, Canada, China, Poland, South Korea, Spain, Taiwan, and Czechia) **Estimated Study Start:** April 2025 **Estimated Primary Completion:** ~13 months per participant (48-week treatment period + 4-week follow-up)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. 18 years of age or older at screening. Adolescents (12 to <18 years of age) are also eligible if permitted by the local IRB/EC and regulatory health authorities. Where these approvals have not been granted, only participants 18 years of age and older will be enrolled. 2. Clinical diagnosis of alopecia areata with no other etiology of hair loss. 3. $\geq 50\%$ hair loss of the scalp, as measured by SALT, without evidence of terminal hair regrowth within the previous 6 months. 4. Current episode of hair loss ≤ 10 years.

4.2 Exclusion Criteria 1. Diseases or conditions other than alopecia areata which affect hair loss (e.g., other types of alopecia, scalp disease, or active systemic diseases). 2. History of severe allergic or anaphylactoid reaction to any kinase inhibitor or known allergy/hypersensitivity to any component (including excipients) of the study intervention. 3. Any psychiatric condition including recent or active suicidal ideation or behavior meeting protocol-defined criteria. 4. General infection history: Systemic infection requiring hospitalization or parenteral therapy within 3 months prior to Day 1, or active acute or chronic infection requiring treatment with oral antibiotics, antivirals, antiparasitics, antiprotozoals, or antifungals within 4 weeks (may rescreen after resolution). Evidence or history of untreated, currently treated, or inadequately treated active or latent *Mycobacterium tuberculosis*. 5. Specific viral infections: History of disseminated herpes zoster or disseminated herpes simplex, or recurrent localized dermatomal herpes zoster. Infected with hepatitis B or hepatitis C viruses (screening required). 6. Other medical conditions: Hearing loss with progression over the previous 5 years, sudden hearing loss, or middle/inner ear disease (e.g., otitis media, cholesteatoma, Meniere's disease, labyrinthitis). Abnormal findings on screening chest imaging (e.g., chest x-ray) including active TB, cardiomyopathy, or malignancy. History of malignancies (with the exception of adequately treated non-metastatic basal/squamous cell skin cancer or cervical carcinoma in situ) or any lymphoproliferative disorders (e.g., EBV-related, lymphoma, leukemia). Significant trauma or major surgery within 1 month of the first dose or in imminent need for surgery. 7. Adolescent participants (12 to <18

years) lacking documented evidence of varicella vaccination (2 doses) or evidence of prior exposure (positive VZV IgG Ab result). 8. Current or prior use of any prohibited medication(s) or vaccine(s). Previous administration of an investigational drug or vaccine within 8 weeks (or 5 half-lives) before the first dose. 9. Diagnostic assessments: Any exclusionary laboratory abnormalities at Screening, or a screening 12-lead ECG demonstrating clinically relevant abnormalities. 10. Investigator site staff directly involved in the study and their family members.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Ritlecitinib 50 mg or 100 mg administered once daily (QD). Participants assigned to 50 mg who are non-responders (SALT ≥ 20) at Week 24 will be re-randomized 2:1 to receive 100 mg or continue 50 mg. **Route of Administration:** Oral (tablets) **Duration of Treatment:** 48 weeks (followed by a 4-week follow-up period).

5.2 Concomitant & Prohibited Therapy Permitted: Standard allowed medications not impacting autoimmune hair loss. **Prohibited:** Participants must be willing to stop all other treatments that they may be taking for alopecia areata.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Up to -12 weeks	Informed Consent, Medical History, Physical Exams, ECG, Chest Imaging (e.g., x-ray), Hearing Test, Blood Tests, SALT assessment
Baseline	Day 0	Randomization, First Dose, Scalp/Eye photographs, Questionnaires
Interim 1	Week 24	Primary Efficacy Assessment (SALT score), Safety Labs, Re-randomization of 50 mg non-responders
Interim 2	Week 36	Efficacy Assessment (comparison vs. Synthetic Placebo), Safety Labs
End of Study	Week 48 / +4 wks	Final Efficacy Assessment, Exit Interview, Safety Follow-Up (Up to 9 total site visits over ~13 months)

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Difference in the percentage of participants with an absolute Severity of Alopecia Tool (SALT) score ≤ 20 between ritlecitinib 100 mg once-daily (QD) versus placebo. **Measurement Method:** Clinical assessment of scalp hair loss compared against an External Placebo Control group derived from prior ritlecitinib trials.

7.2 Secondary & Exploratory Endpoints * Percentage of participants with an absolute SALT score ≤ 20 (evaluated at

various points including Week 36 and Week 48). * Change from baseline in SALT score. * **EU Only:** Percentage of participants with Patient Global Impression of Change (PGI-C) response, defined as a score of "moderately improved" or "greatly improved".

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Routine collection via site visits, physical examinations, hearing tests, electrocardiograms (ECG), blood tests, and participant questionnaires. **Serious Adverse Events (SAEs):** Standard 24-hour reporting timelines for severe instances (e.g., severe allergic reactions, serious infections, cardiovascular events). **Data Monitoring Committee (DMC):** External independent safety committee in place to monitor the dose-up safety and overall study risk.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for efficacy; Safety Set for AE/SAE evaluations. **Sample Size Justification:** \$N \approx 550\$ participants powered to detect superiority of 100 mg QD vs 50 mg QD and external controls. **Handling of Missing Data / Novel Models:** Efficacy will be analyzed under a fully Bayesian framework using Bayesian beta-binomial models. A Synthetic Placebo Control group utilizes longitudinal statistical modeling to extrapolate outcomes over 36 weeks.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Continuous remote data capture and rigorous onsite validation across up to 9 site visits per participant. **Audit Trail:** eCRF logging of SALT score evaluations, central reading of scalp photography, and continuous safety audits.

11. Study Identifiers & Governance

Primary ID: B7981094 **Registry Number:** NCT06873945 **Posting ID:** 674030387309581743 **Sponsor/Lead Organization:** Pfizer **Study Title:** A Study of 2 Doses of Ritlecitinib in People 12 Years of Age and Older With Alopecia Areata **Official Regulatory Title:** A PHASE 3, EXTERNAL AND SYNTHETIC PLACEBO-CONTROLLED RANDOMIZED STUDY WITH DOSE-UP FOR NON-RESPONDERS TO INVESTIGATE SAFETY AND EFFICACY OF RITLECITINIB 50 MG AND 100 MG ONCE DAILY IN ADULT AND ADOLESCENT PARTICIPANTS 12 YEARS OF AGE AND OLDER WITH ALOPECIA AREATA **Trial Status:** Recruiting

12. Clinical Framework (Alopecia Areata Specific)

Primary Condition: Severe Alopecia Areata (AA) **Diagnosis Threshold:** $\geq 50\%$ hair loss of the scalp, as measured by SALT **Medical Methodology:** Targeted Kinase Inhibitor (JAK3/TEC family) therapy **Optimization Target:** Absolute SALT score ≤ 20 (i.e., $\leq 20\%$ scalp hair loss)

13. Study Procedures & Methodology

Management Pathway: Oral ritlecitinib therapy (50 mg or 100 mg daily) with dose escalation framework for non-responders. **Education Protocol (HCP):** Site initiation visits and standardized training on SALT measurement, photography protocols, and synthetic control methodology. **Education Protocol (Patient):** Informed consent (including caregiver consent for adolescents), treatment compliance, and washout of prior AA treatments. **Quality Audit Mechanism:** Blinded assessor review for SALT scoring and Bayesian modeling data integrity tracking.

14. Observation & Follow-Up Schedule

Total Duration: ~13 months (48-week treatment + 4-week follow-up) **Onsite Visit Cadence:** Up to 9 visits (Screening, Baseline, and evaluation points including Weeks 24, 36, and 48) **Remote Monitoring Frequency:** Standard protocol-defined intervals. **Checklist Review Interval:** Continuous AE monitoring, laboratory panel checks, and ECG/hearing reassessments as scheduled.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				